



world frame (plane of motion)



gravity in-plane: $[0, -g, 0]$

COM of dipper+handle assembly

$M_d = 54,300 \text{ kg} \cdot I_{zz} = 287,900 \text{ kg} \cdot \text{m}^2$ (Rasuli Table II)

$L_{\text{COG}} = 2.82 \text{ m}$ (Rasuli Table II)

dipper handle (slides through saddle)

crowd DOF: $\dot{d}_4, F_4$

saddle block (collar)

$\tau_3, \dot{q}_3$

$q_3 = 45^\circ \pm 15^\circ$

saddle pivot — revolute joint

(boom = passive structure $\rightarrow$ ground)

prismatic (crowd extension)
 $8.75 \pm 2 \text{ m}$
lever arm $r = d_4 - 1.32 \text{ m}$ (TIME-VARYING)

1.5 m rigid offset: pivot $\rightarrow$ handle frame (our construction)

Stage-1 kinematic skeleton — 2 DOF:

q_3 (saddle rotation) + d_4 (crowd extension).

Swing & hoist rope enter at M5-M6. Tilted-g planar model.